IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE PART IV: MEng and ACGI

ANALOGUE SIGNAL PROCESSING

Friday, 15 May 2026, 10:00 AM

Time allowed: 3:00 hours

There are SIX questions on this paper.

Answer FOUR questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : C. Papavassiliou

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Special instructions for students

Unless otherwise stated the following parameters have the following definitions:

V_{DS} *: Drain Source Voltage.*

V_{GS} *: Gate Source Voltage.*

V_{TH} *: Threshold Voltage.*

U_t *: Thermal Voltage.*

n *: Weak Inversion Slope factor.*

g_m*: Transconductance.*

1.

a) State two advantages and two disadvantages of encoding analogue signals in time rather than amplitude.

[4]

b)

i. Draw a circuit that encodes a current mode signal in a domain that is continuous in time but discrete in amplitude.

[4]

ii. Explain its operation using both equations of the relationship between the input and output signals as well as time domain plots of the amplitude of an encoded analogue signal.

[4]

c) For the cascade of amplifiers shown in Figure 1.1, derive the output referred noise and explain what factors can be optimized to improve the signal to noise ratio.

[8]

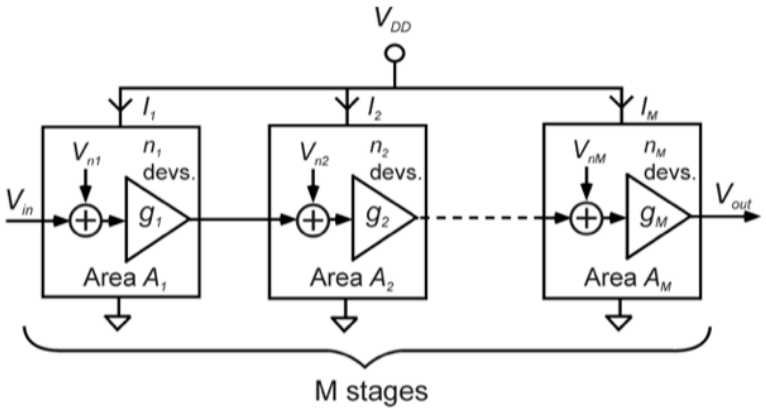


Figure 1.1: Cascade of amplifiers

2. This question concerns translinear Circuits.

a) Give two limitations when using MOSFETs in translinear circuits.

[2]

b) Figure 2.1 shows a translinear circuit.

i. Derive the function of the circuit using the translinear principle. I_z is the output and I_x is the input.

[2]

ii. Show how, with the addition of a few transistors, the circuit can be modified to calculate the function:

$$I_Z = \frac{(I_X)^4}{I_0^3} - I_x$$

[4]

iii. A capacitor is added between the point A on the circuit in Figure 2.1 and ground. Derive the new transfer function of the circuit stating the gain and time constant. The final equation of the transfer function should be represented in the s-domain.

[4]

c) Figure 2.2 shows a translinear circuit capable of computing the trigonometric function sin(πy) through the approximation $\frac{y-y^3}{1+y^2}$.

i. Express the currents I_3 and I_4 in terms of I and I_X.

[4]

ii. Express the differntial output current I_2-I_1 and show that when I_X = yI then I_2-$I_1 = I\frac{y-y^3}{1+y^2}$

[4]

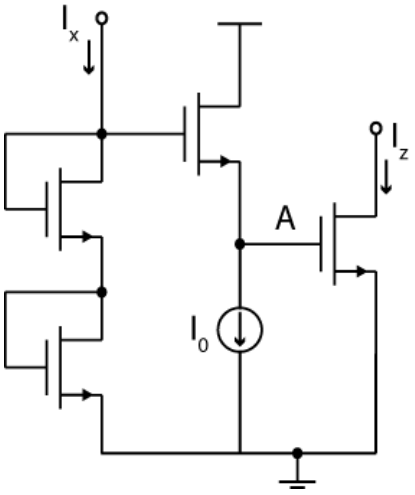


Figure 2.1

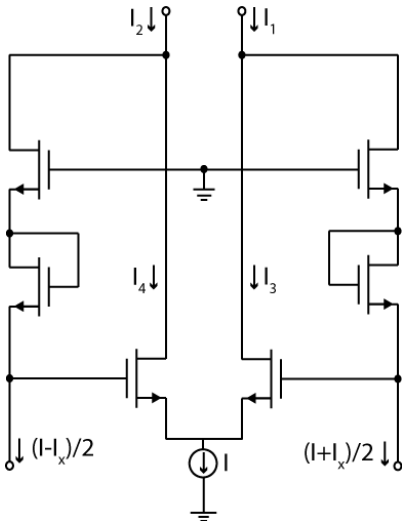


Figure 2.2

3.

a) Explain the principle of companding, illustrating how the dynamic range changes through the signal processing chain.

[4]

b) Figure 3.1 shows a block diagram of a current-mode companding integrator, with an input X and an output Y. f() and g() represent non-linear functions which compress and expand the signal such that the input-output relationship of the integrator is linear and given by:

$$Y = \tau \int X \cdot dt$$

Given that the expansive function is defined as $I_{out} = g(V_c) = I_0 \sinh(V_c/nU_t)$, derive the function f(x) such that linear integration is achieved. All constants should be grouped to represent a current I_1, with the function f(x) composed of just I_{in}, I_1 and I_{cosh}, whereby $I_{cosh} = I_0 \cosh(V_c/nU_t)$.

[6]

c) The transfer function of a biquad filter is defined in state space representation by the following equations:

$$\dot{X}_1 = \frac{-\omega_0}{Q} X_1 - \omega_0 X_2 + \omega_0 U_1$$

$$\dot{X}_2 = \omega_0 X_1 - \omega_0 U_2$$

$$Y = X_1$$

whereby Y is the output and U_1 and U_2 are the inputs and X_1 and X_2 are the states.

i. By using the mappings below show how these linear equations can be mapped to non-linear log-domain design equations. State any assumptions you make.

$$X_1 = I_1 \exp\left(\frac{V_1}{nU_t}\right) \qquad X_2 = I_2 \exp\left(\frac{V_2}{nU_t}\right) \qquad U_2 = I_{U2} \exp\left(\frac{V_{U2}}{nU_t}\right)$$

[5]

ii. With these log-domain design equations, sketch a schematic of the final log domain filter using weak inversion MOS transistors.

[5]

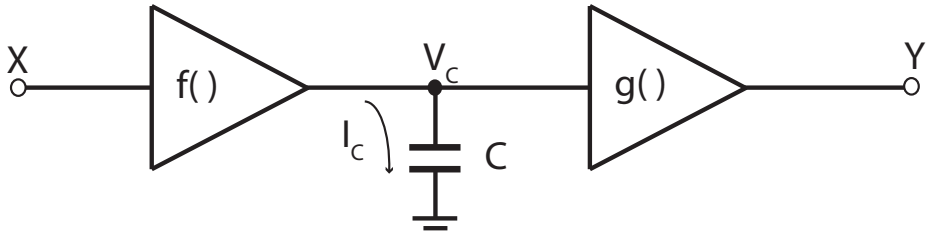


Figure 3.1

4.

a) Figure 4.1 shows a conventional second-generation positive current conveyor, CCII+.

i. Explain its operation principle and describe how impedances at each of its three ports, X, Y, Z, differ from a standard operational amplifier.

[4]

ii. Draw the circuit of a CMOS implementation of a bi-directional CCII+ with high output impedance.

[4]

iii. Using the current conveyor in Figure 4.1, draw the schematics of a current mode integrator and a voltage mode integrator, each with a time constant, τ =1s. You may assume R = 100KΩ.

[4]

b) Figure 4.2 shows a voltage mode Sallen key filter. Using the adjoint principle transform this to it's current mode equivalent using CCII+ blocks.

[4]

c) Sketch a current mode circuit that can implement the following hyperbolic functions:

$$I_1 - I_2 = I_{DC} \sinh\left(\frac{V_1 - V_2}{nU_t}\right)$$

$$I_1 + I_2 = I_{DC} \cosh\left(\frac{V_1 - V_2}{nU_t}\right)$$

[4]

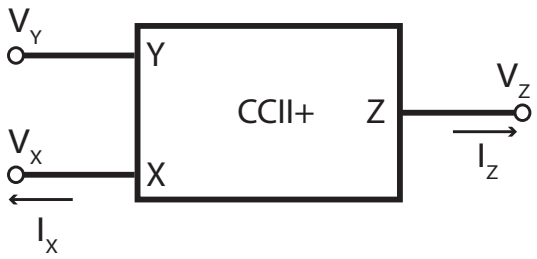


Figure 4.1

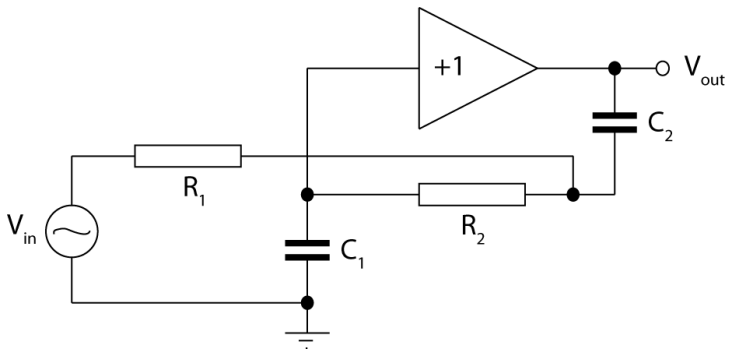


Figure 4.2

5. This question relates to discreet time integrators.

a) Figure 5.1 shows a discrete time RC integrator.

i. Derive the time constant of integration.

[2]

ii. Assuming $C_1 = C_2$, select a suitable clock frequency to give a time constant $\tau = 1\mu s$.

[2]

iii. For the circuit Figure 5.1, when CK goes low all of the channel charge, Q_1, of transistor M_1 is injected into C_1. This causes an error voltage ΔV_{C1} in C_1 defined by:

$$\Delta V_{C1} = \frac{Q_1}{C_1}$$

Then transistor M_2 turns ON. The channel charge of M_2 is split equally between C_1 and C_2. Estimate the total amount of charge injection and therefore error voltage ΔV_{C2} from switches M_1 and M_2 which appears on the output capacitor C_2 after CK goes high again, that is after one integration cycle.

[4]

b) Figure 5.2 shows an autozeroing switched capacitor integrator capable of removing any offset in the amplifier. Derive the output voltage of the integrator during phase Φ_1 and Φ_2 indicating how the offset is removed by showing the charge on each capacitor during each phase.

[5]

c) Figure 5.3 shows a schematic of switched current integrator.

i. Derive the transfer function of the switched current integrator and show that in the z-domain this may be represented as H(z) = A/(z-1).

[5]

ii. Explain one limitation of the switched current technique that doesn't exist in its voltage mode equivalent.

[2]

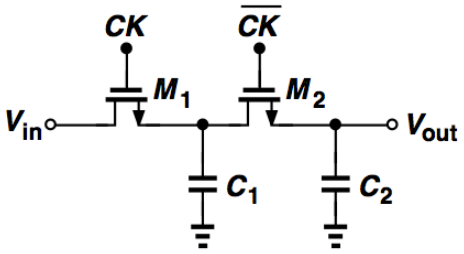


Figure 5.1

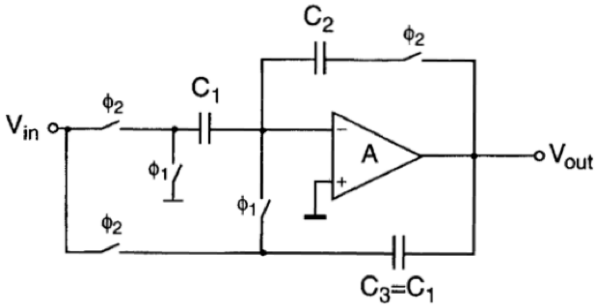


Figure 5.2

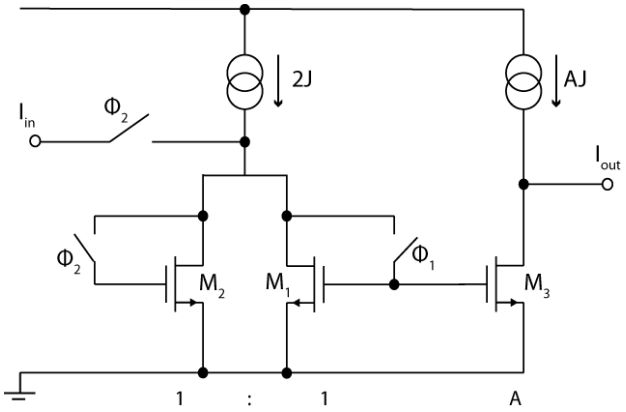


Figure 5.3

6. Figure 6.1 shows a biomedical system used to measure the similarity between two EEG channels by recording action potentials from the human brain. The front end consists of a differential operational amplifier (OA) connected to a voltage-to-current (V/I) converter. This then connects to a current rectifier followed by a peak detector that detects the envelope of the converted current I_{out1}. The similarity between the envelopes of the two channels, I_{env1} and I_{env2}, is detected using a current mode correlator to derive an output signal I_{corr}.

   a) Explain one method to make the input signals V_{sig1} and V_{sig2} immune to 1/f noise in the operational amplifier and draw a block diagram of this method illustrating how the frequency spectrum of the signal changes during each stage.

   [5]

   b) Draw a schematic of a suitable front end for this system using the method described in 6.a. You may use the OA symbol as in Figure 6.1 and do not need to show the schematics of this amplifier.

   [5]

   c) Figure 6.2 shows a suitable circuit that can be used as a current mode peak detector. Explain its principle of operation.

   [5]

   d) The system in Figure 6.1 uses a V/I converter with improved linearity over a conventional differential pair. Sketch suitable schematics for the V/I converter and the current correlator used in this system.

   [5]

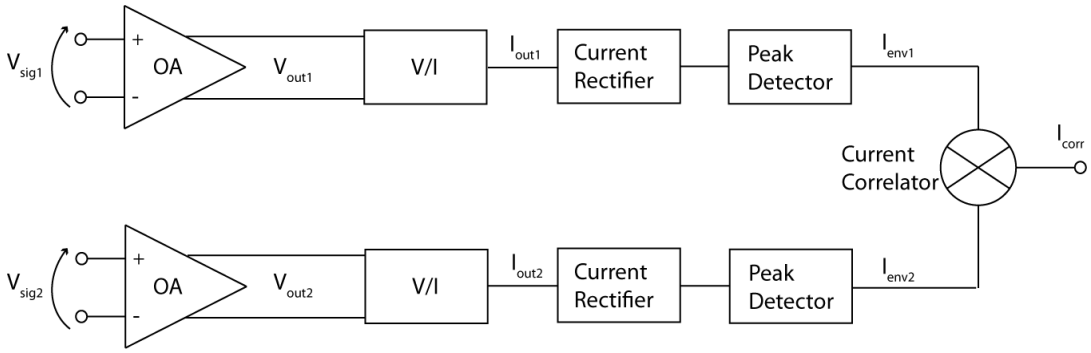


Figure 6.1

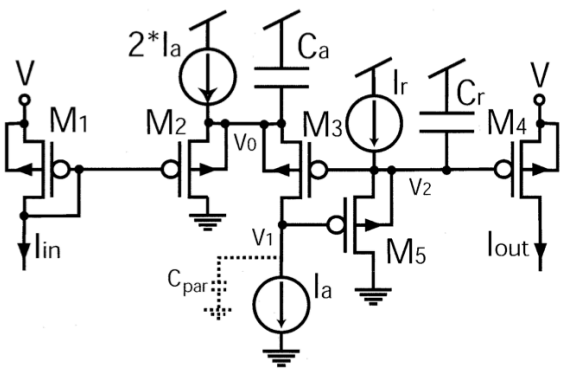


Figure 6.2